操作文档

配置环境 gedit ~/.bashrc 文档末尾加入 source /opt/ros/noetic/setup.bash source ~/xxxxx/devel/setup.bash xxxxx为你的文件路径

1. 启动仿真环境

用途：启动UR5机械臂与MoveIt仿真环境

```
roslaunch ur5_moveit_config demo.launch
```

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2. 六点点位控制

用途：执行六个目标点位运动

```
rosrun arm_control_demo six_points
```

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3. 笛卡尔直线轨迹

用途：执行末端直线运动

```
rosrun arm_control_demo cartesian_line
```

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4. 椭圆轨迹

用途：执行椭圆轨迹运动

```
rosrun arm_control_demo ellipse_path
```

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5. 8字轨迹

用途：执行8字轨迹运动

```
rosrun arm_control_demo eight_path
```

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6. 虚拟画板系统

终端1

用途：启动UR5仿真环境

```
roslaunch ur5_moveit_config demo.launch
```

终端2

用途：启动Qt虚拟画板

```
rosrun virtual_canvas_gui virtual_canvas_gui
```

终端3

用途：启动轨迹执行器

```
rosrun canvas_executor canvas_executor
```

使用流程

1. 启动终端1
2. 启动终端2
3. 启动终端3
4. 在Qt画板中绘制轨迹
5. 点击“执行”
6. 观察机械臂运动
7. 点击“清空”重新绘制

命令速查

```
roslaunch ur5_moveit_config demo.launch

rosrun arm_control_demo six_points

rosrun arm_control_demo cartesian_line

rosrun arm_control_demo ellipse_path

rosrun arm_control_demo eight_path
```

```
rosrun virtual_canvas_gui virtual_canvas_gui

rosrun canvas_executor canvas_executor
```